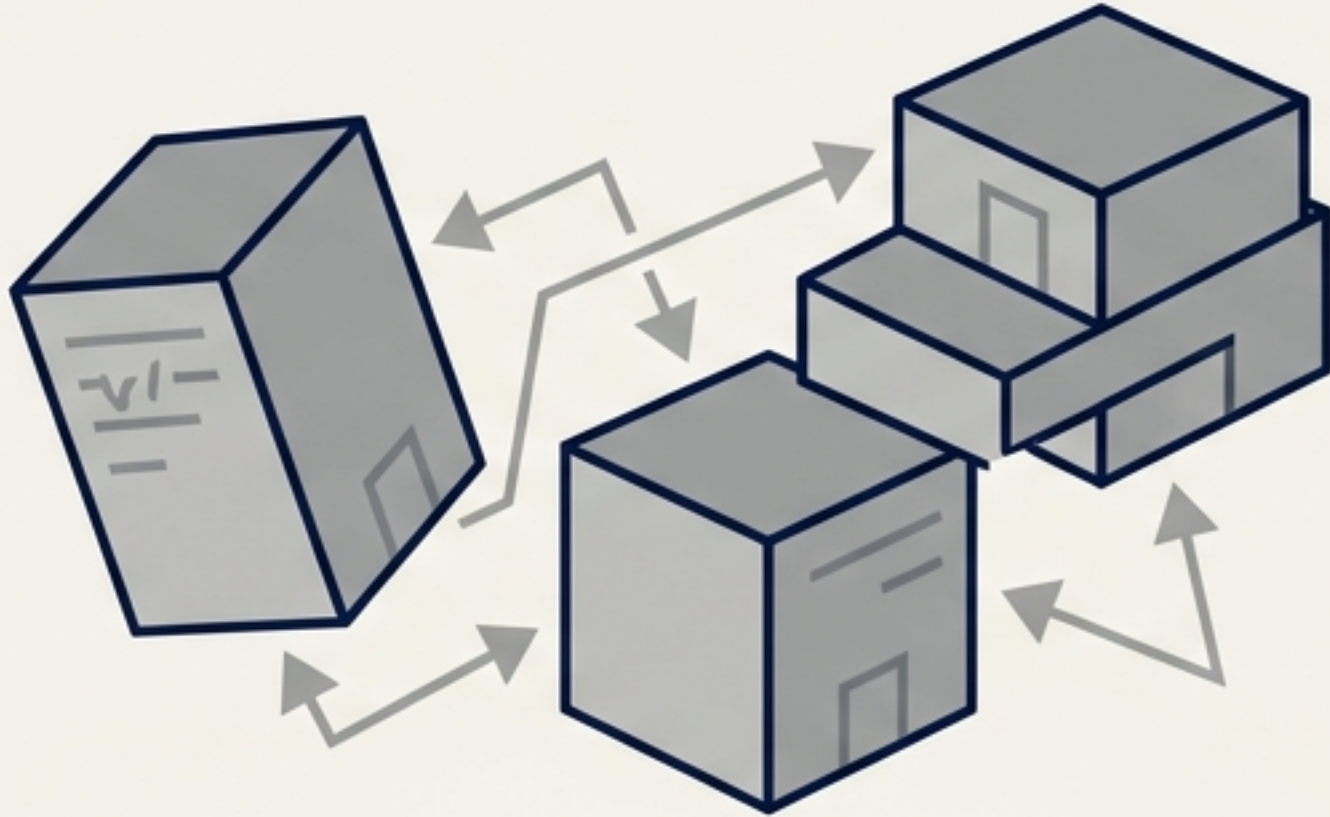


E-Commerce Analytics: Strategic Insights & Infrastructure

Moving from data silos to
actionable intelligence.

The Cost of Fragmented Data

Current State: Data Silos & Blind Spots



Unable to accurately track total revenue without manual effort.

No visibility into regional bottlenecks.

Disconnect between product performance and logistics.

Target State: A Single Source of Truth



What is our true revenue and average order value?

Which categories drive the highest margin?

Where are our delivery failures costing us customers?

The End-to-End Analytics Workflow



Extract & Clean

Python / Pandas

Process raw datasets via Kaggle API. Standardise customer, product, and order records.



Model & Store

SQL / SQLite

Load cleaned data into structured staging tables. Build relational architecture.

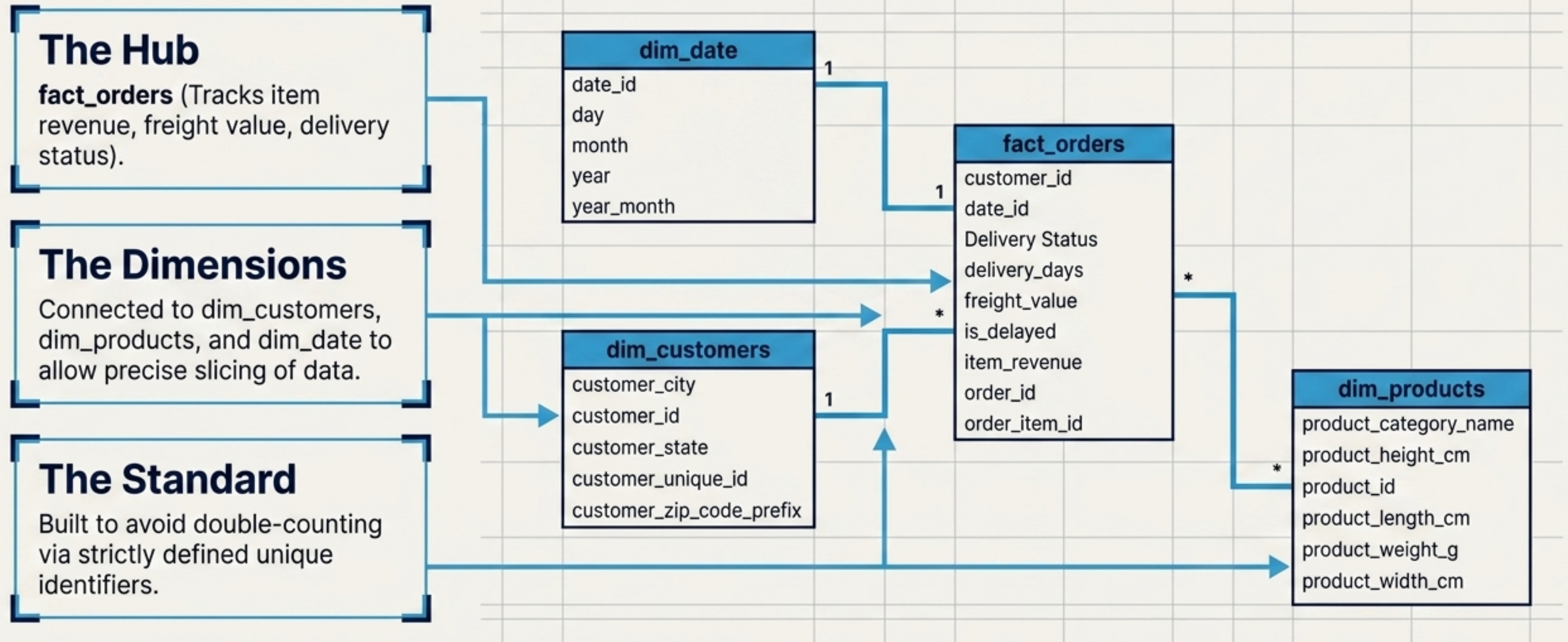


Visualise & Decide

Power BI

Translate raw metrics into executive dashboards for strategic monitoring.

The Engine: SQL Star Schema



Sales Overview: The Top-Line Picture

15.42M

Total Revenue

96K

Completed Orders

\$159.83

Average Order Value (AOV)

Baseline operational speed sits at 12.01 average delivery days.



Product Performance: Drivers of Revenue

Tier 1 Category

Health & Beauty (beleza_saude)

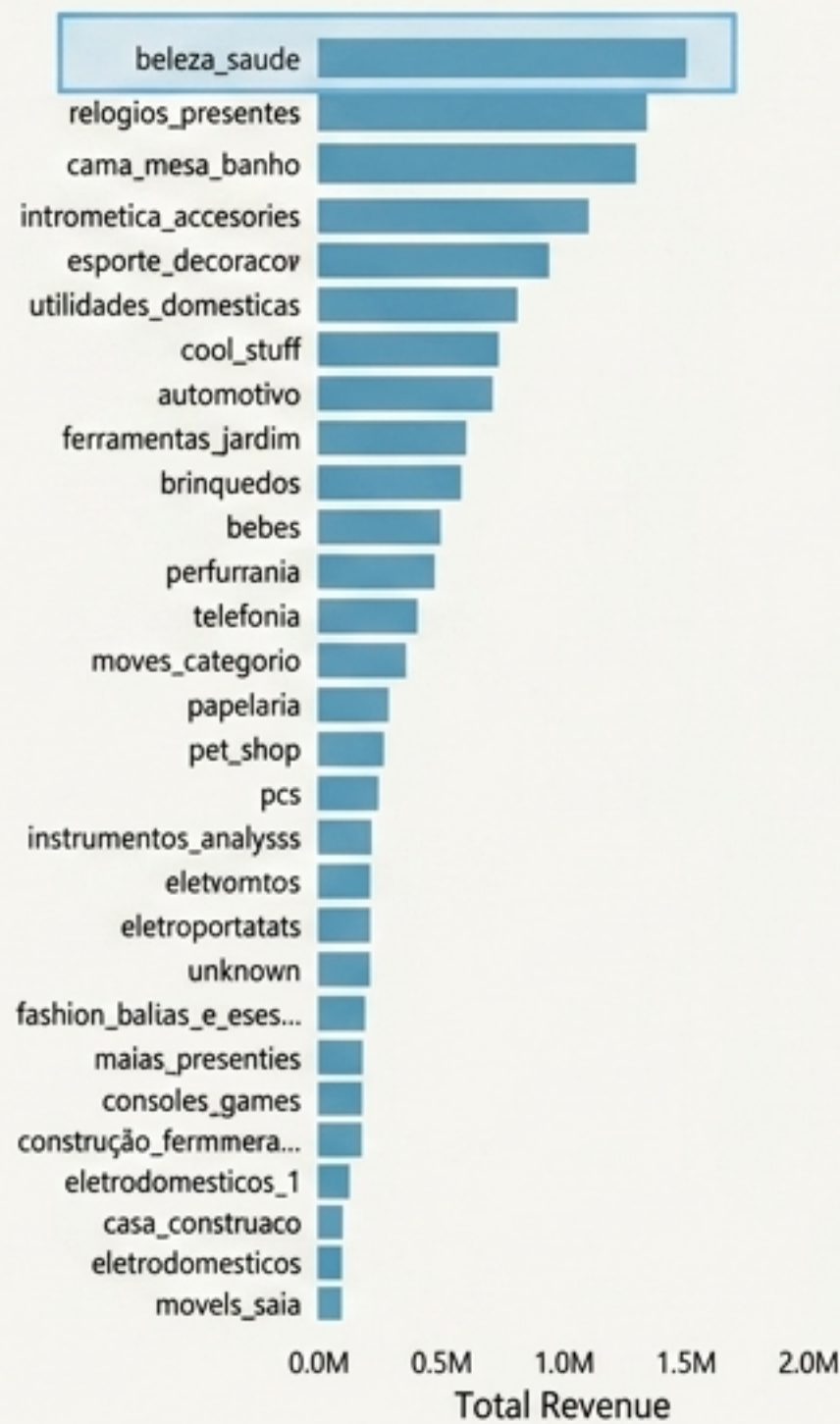
Runner-Up Categories

Watches/Gifts (relogios_presentes)
Bed/Bath (cama_mesa_banho)

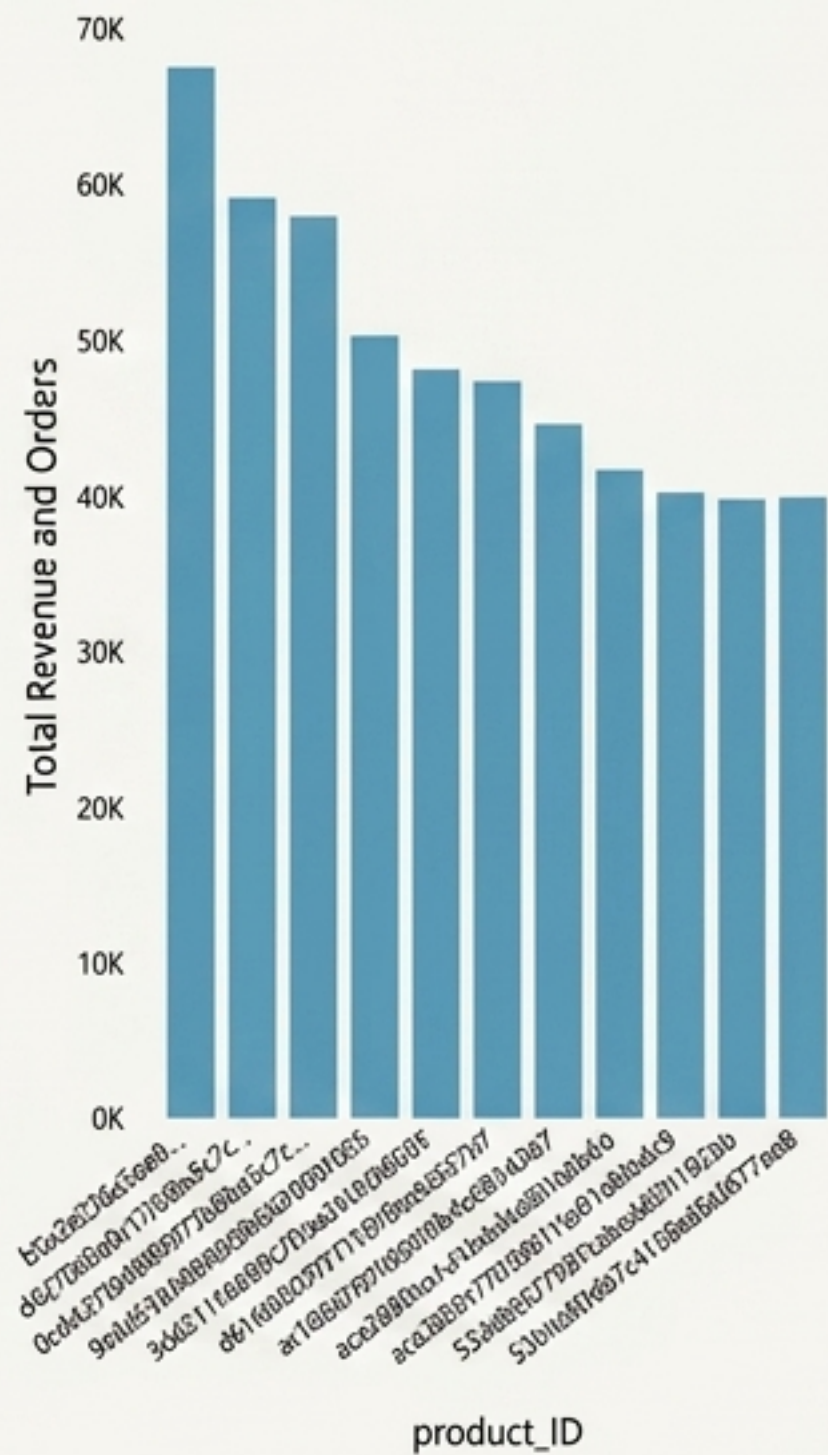
Key Insight

High revenue concentration. Top individual products are generating between 40K and 67K each in revenue, significantly outpacing the long tail.

Revenue by Product Category



Top Products by Revenue



Regional Performance: Market Concentration



Market Leaders

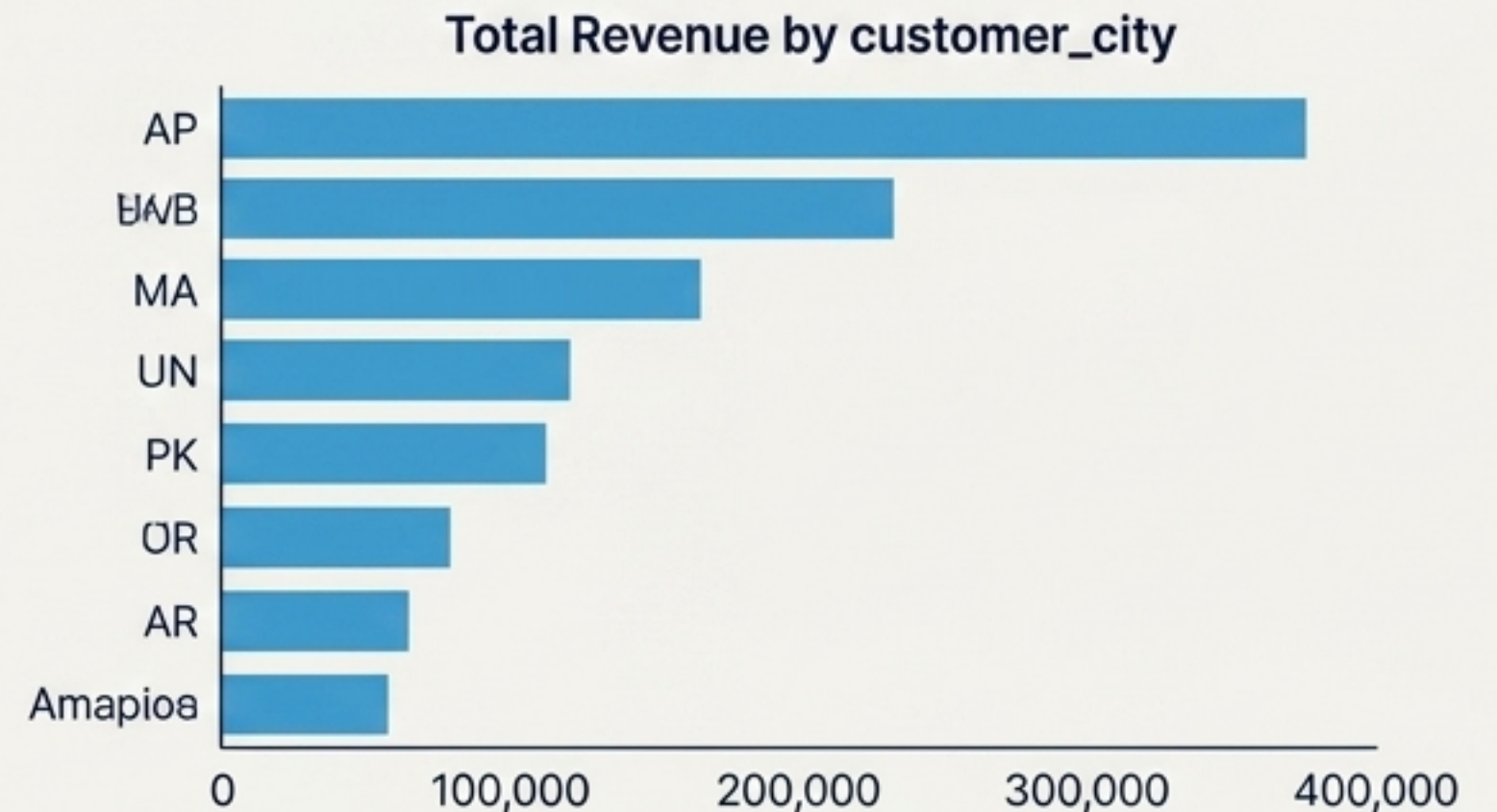
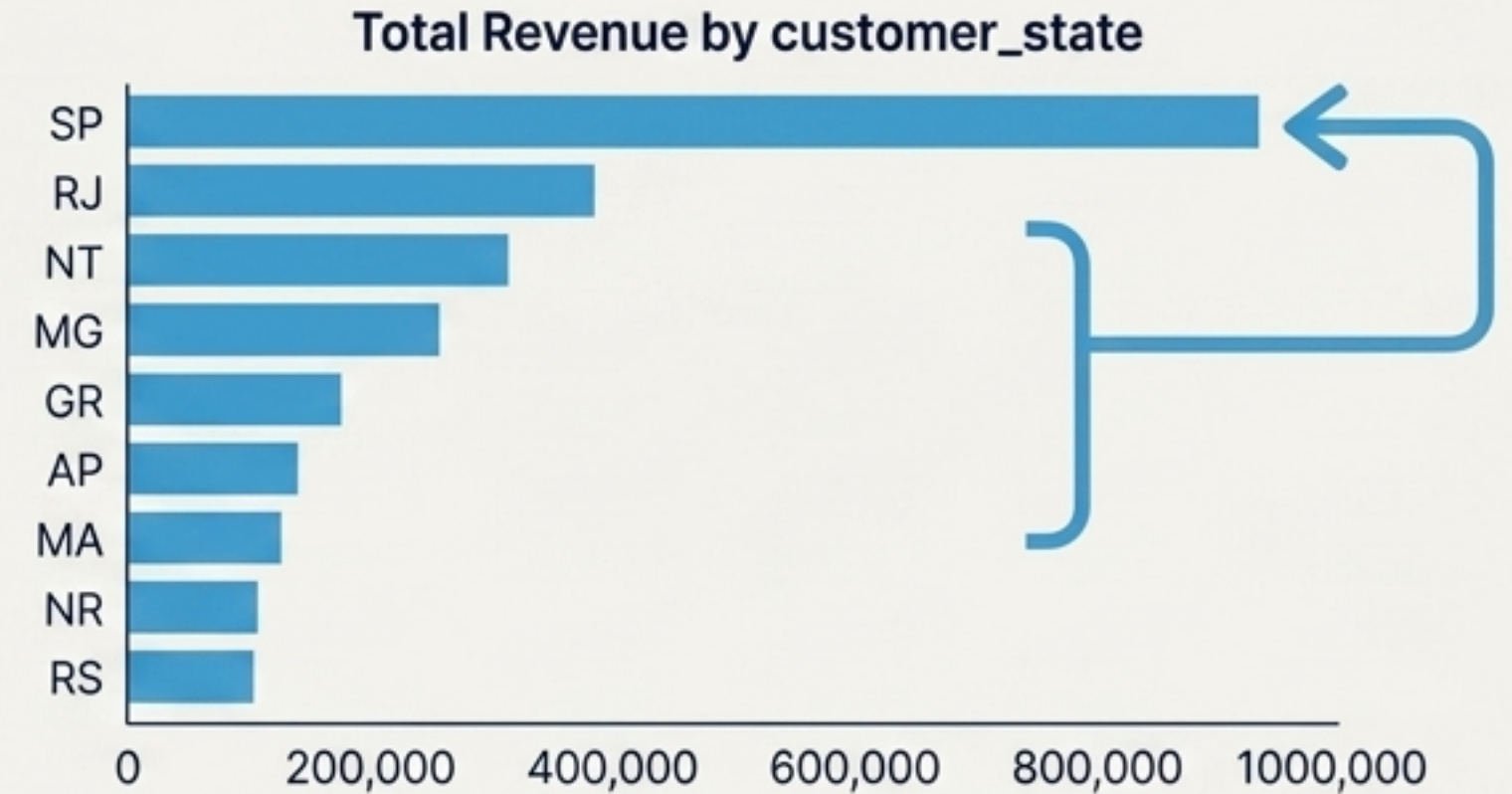
São Paulo (SP) holds massive dominance, supported by Rio de Janeiro (RJ) and Minas Gerais (MG).



Underperforming Regions

Roraima (RR), Amapá (AP), and Acre (AC) sit at the bottom of the revenue tail.

Strategic Implication: Revenue is highly concentrated in the Southeast corridor.



Delivery Performance: The Customer Experience

91.89%

On-Time Delivery Rate (89K Orders)

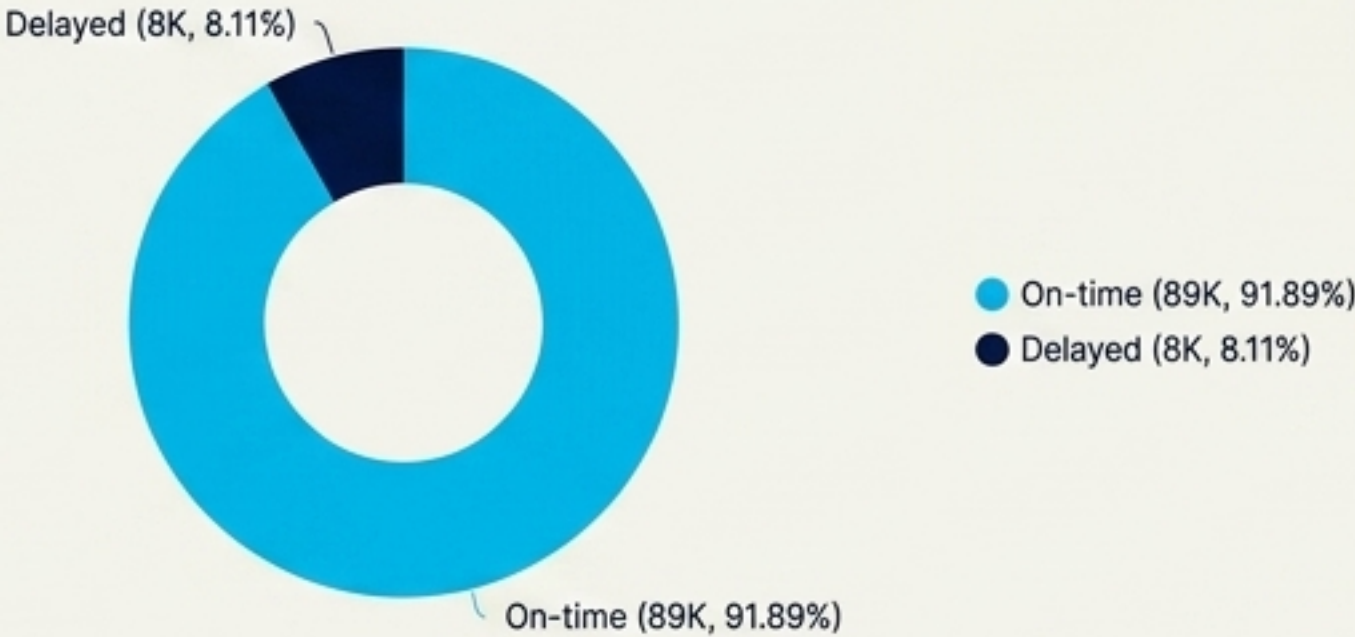
Average Speed: 12.01 Days.

8.11%

Delayed Delivery Rate (8K Orders)

Insight: While the baseline success rate is strong, 8,000 delayed orders represent a significant volume of compromised customer experiences.

Delayed vs On-time



Delivery Days Trend



Strategic Recommendations: The Path Forward



Pillar 1: Product Optimisation

Double-down marketing spend on Health & Beauty (beleza_saude).

Ensure priority inventory availability for top-tier SKUs generating 40K+ revenue.



Pillar 2: Regional Expansion

Secure dominance in the SP/RJ/MG corridor with loyalty campaigns.

Conduct pricing and delivery audits for low-performing regions (RR, AP, AC).



Pillar 3: Logistics Efficiency

Audit carriers associated with the 8,000 delayed orders.

Renegotiate logistics partnerships to protect customer retention.

Conclusion: Infrastructure Built for Growth

Business Intelligence Achieved

Validated 15.4M Revenue and 96K total orders.

Identified specific product and regional revenue drivers.

Quantified logistical friction points (8.11% delay rate).

Scalable Foundation Built

Automated Python/SQL data extraction pipeline.

Robust Star Schema eliminating data silos.

Live Power BI dashboards ready for future data integration.